

Electrothermal Manipulation of Abrikosov Vortices via Localized Heating

Su Youyou
NTU Singapore
Summer Research 2026 @ TDLI

Abstract

Abrikosov vortices in topological superconductors may host Majorana zero modes, whose manipulation is essential for future quantum computation. However, direct control of Majorana states remains experimentally challenging. In this project, we investigate an alternative approach: controlling vortex motion through localized electrothermal effects. A two-dimensional overdamped vortex dynamics model is developed, incorporating thermal driving force, vortex-vortex interaction and pinning effects. Numerical simulations demonstrate vortex capture, controlled transfer between electrodes, and programmable circular vortex motion using sequentially activated hotspots.

Motivation

Topological superconductors may host Majorana zero modes (MZMs) inside Abrikosov vortex cores. Future quantum computation requires controlled vortex manipulation for possible braiding operations. However, directly manipulating and observing Majorana modes remains experimentally challenging.

Research Question

This project studies whether localized electrothermal effects can control vortex motion. The main questions are:

- 1 Can a localized hotspot capture a vortex?
- 2 Can two hotspots transfer a vortex?
- 3 Can programmable hotspots guide vortex motion?

Physical Picture

Bottom Electrode



Joule Heating



∇T



Vortex Motion

A local temperature gradient produces an effective force on vortices.

Physical Model

The vortex dynamics is driven by the temperature gradient generated by localized electrothermal heating.

Temperature field

The local temperature distribution is approximated by Gaussian hotspots:

$$T(\mathbf{r}, t) = T_0 + \sum_k \Delta T_k(t) \exp\left(-\frac{|\mathbf{r} - \mathbf{R}_k|^2}{2\sigma_T^2}\right)$$

where:

- T_0 : background temperature
- $\Delta T_k(t)$: heating amplitude of hotspot k
- $\mathbf{R}_k$: hotspot position
- σ_T : hotspot width

Vortex dynamics

A local temperature gradient produces an effective thermal driving force:

$$\mathbf{f}_T = \alpha_T \nabla T$$

The vortex motion is assumed to be overdamped:

$$\eta \frac{d\mathbf{r}}{dt} = \mathbf{f}_T + \mathbf{f}_{pin} + \mathbf{f}_{vv}$$

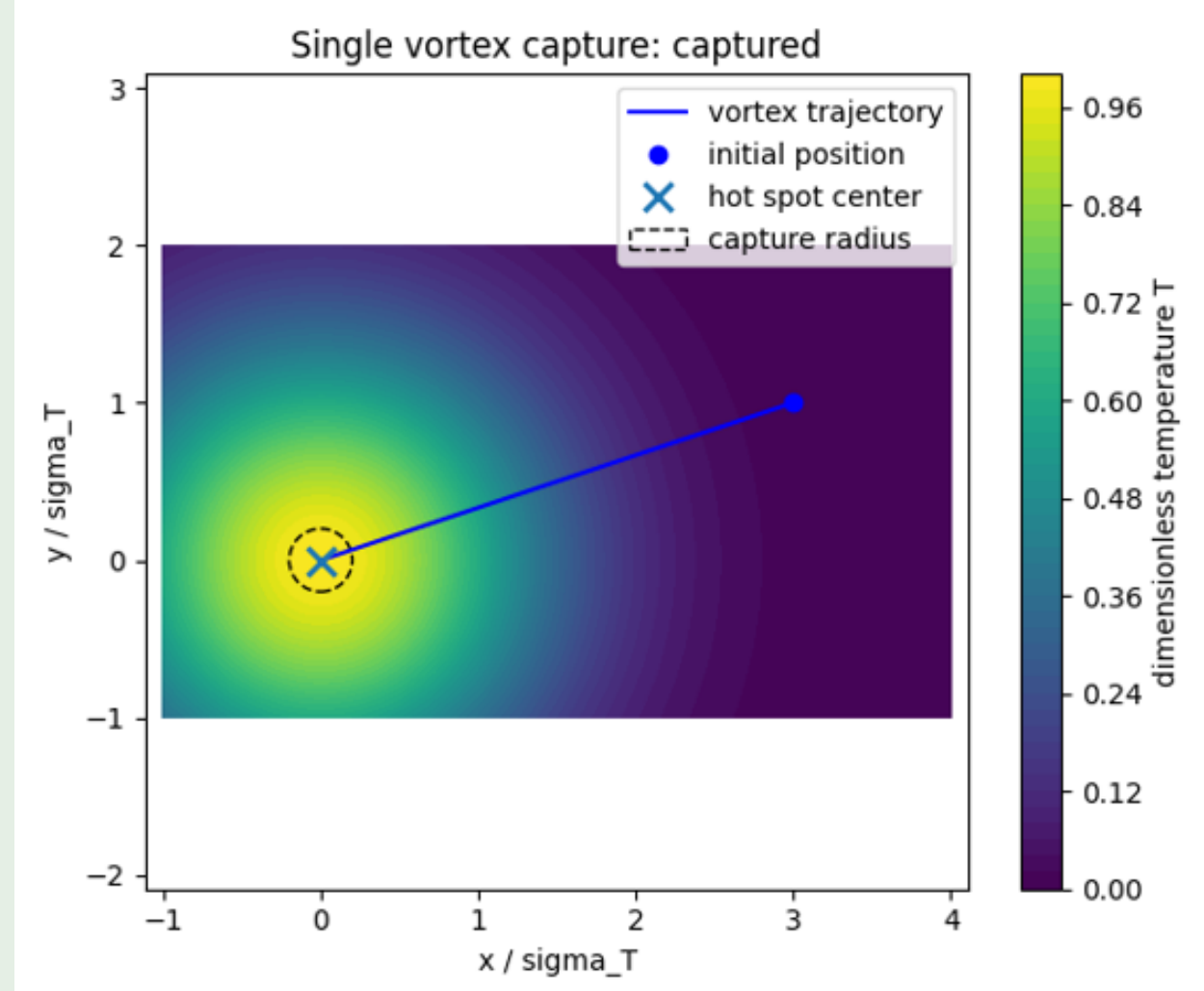
where $\mathbf{f}_{pin}$ and $\mathbf{f}_{vv}$ represent pinning and vortex-vortex interaction forces.

Combining the above equations gives the governing equation:

$$\frac{\Phi_0 B_{c2}}{\rho_n} \frac{d\mathbf{r}}{dt} = \frac{\Phi_0^2}{4\pi\mu_0\lambda_0^2 T_c} \ln\left(\frac{\lambda_0}{\xi_0}\right) \nabla T + \mathbf{f}_{pin} + \mathbf{f}_{vv}$$

The inertial term is neglected because vortex motion occurs in the strongly dissipative regime.

Single Hotspot Capture

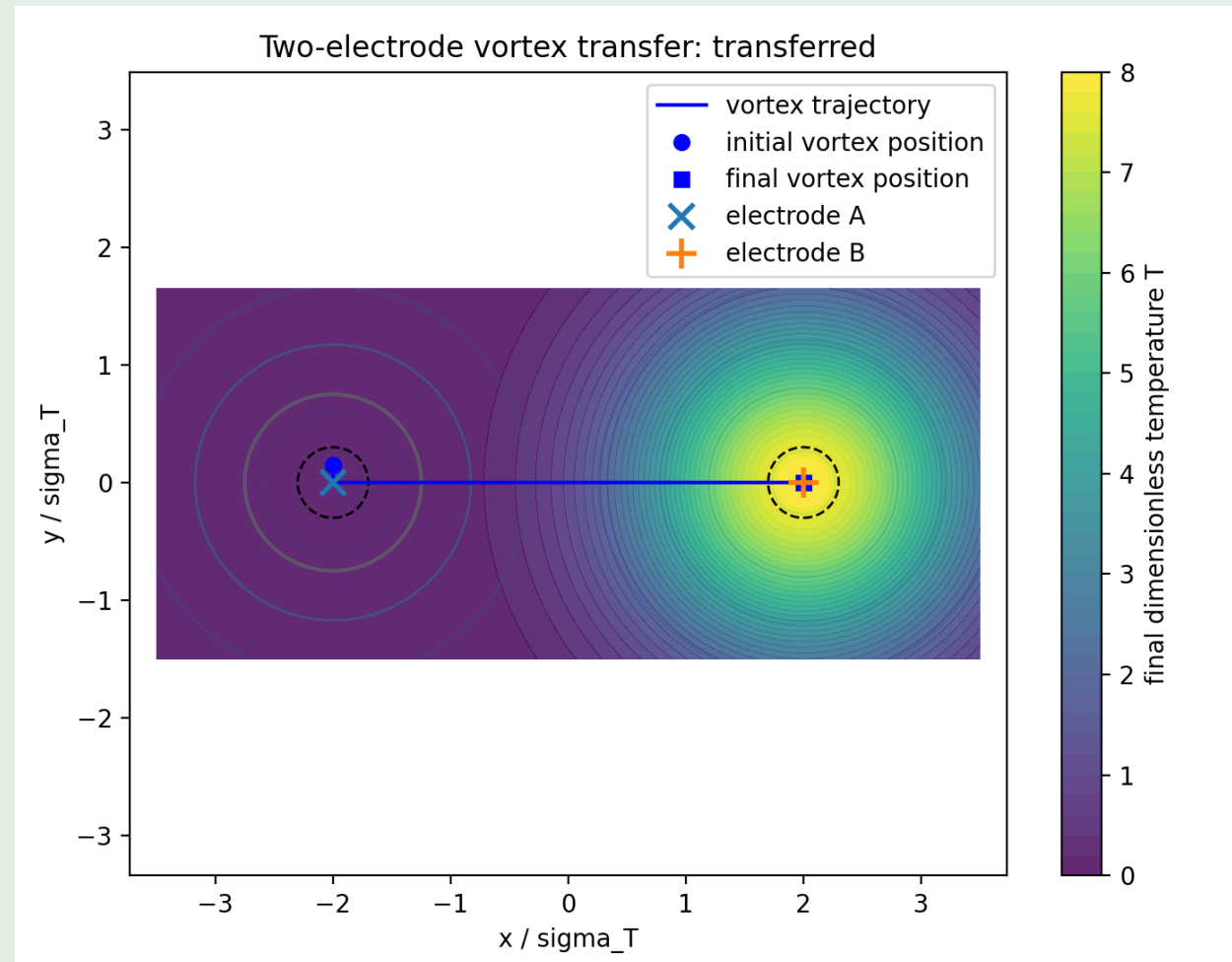


A vortex initially far from the hotspot is attracted toward the high temperature region.

Result

- A single hotspot can capture a vortex.
- Increasing heating strength enlarges the capture radius.

Vortex Transfer Between Electrodes



Two hotspots are controlled by:

$$\Delta T_A(t) \downarrow$$

$$\Delta T_B(t) \uparrow$$

The vortex follows the moving temperature landscape.

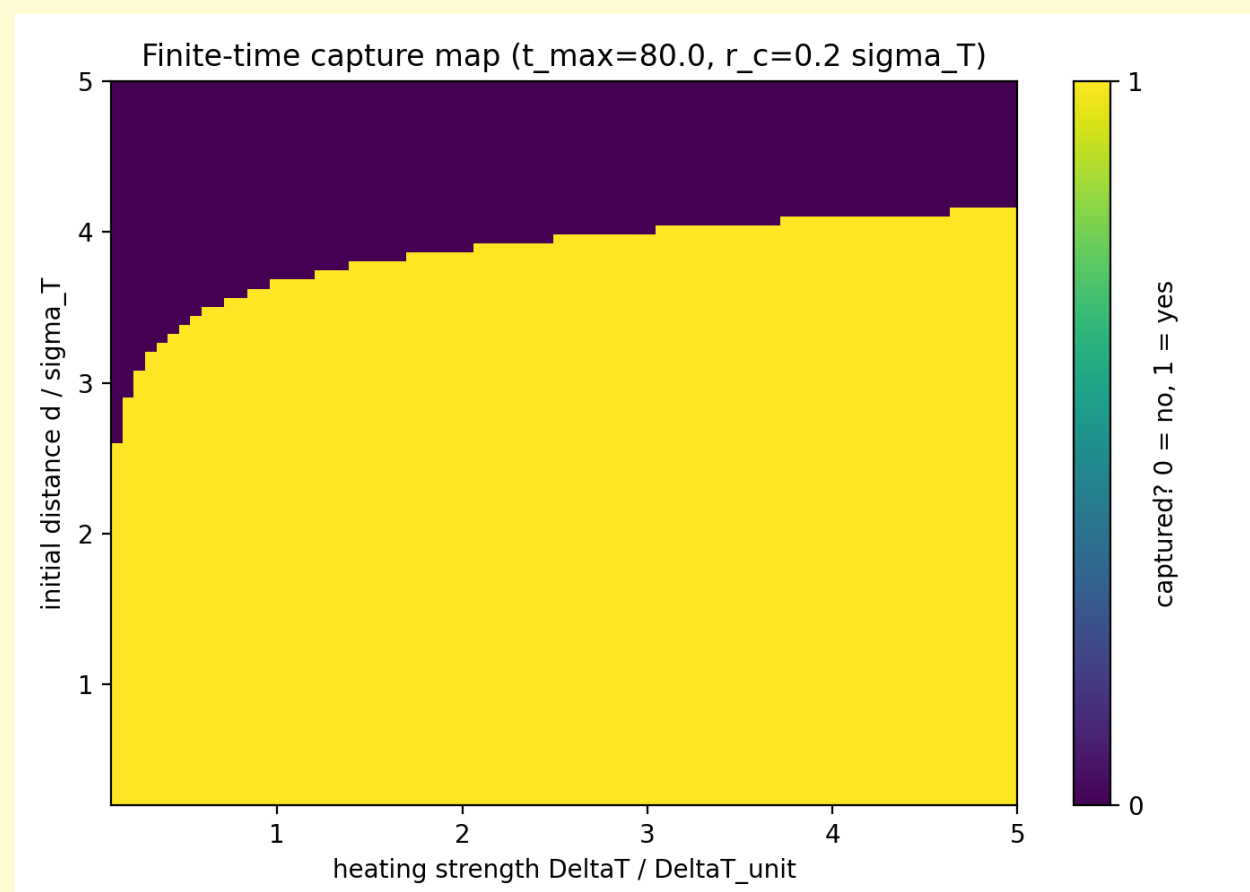
Result:

Controlled vortex transfer is achieved.

Transfer Parameter Dependence

Successful vortex transfer depends on:

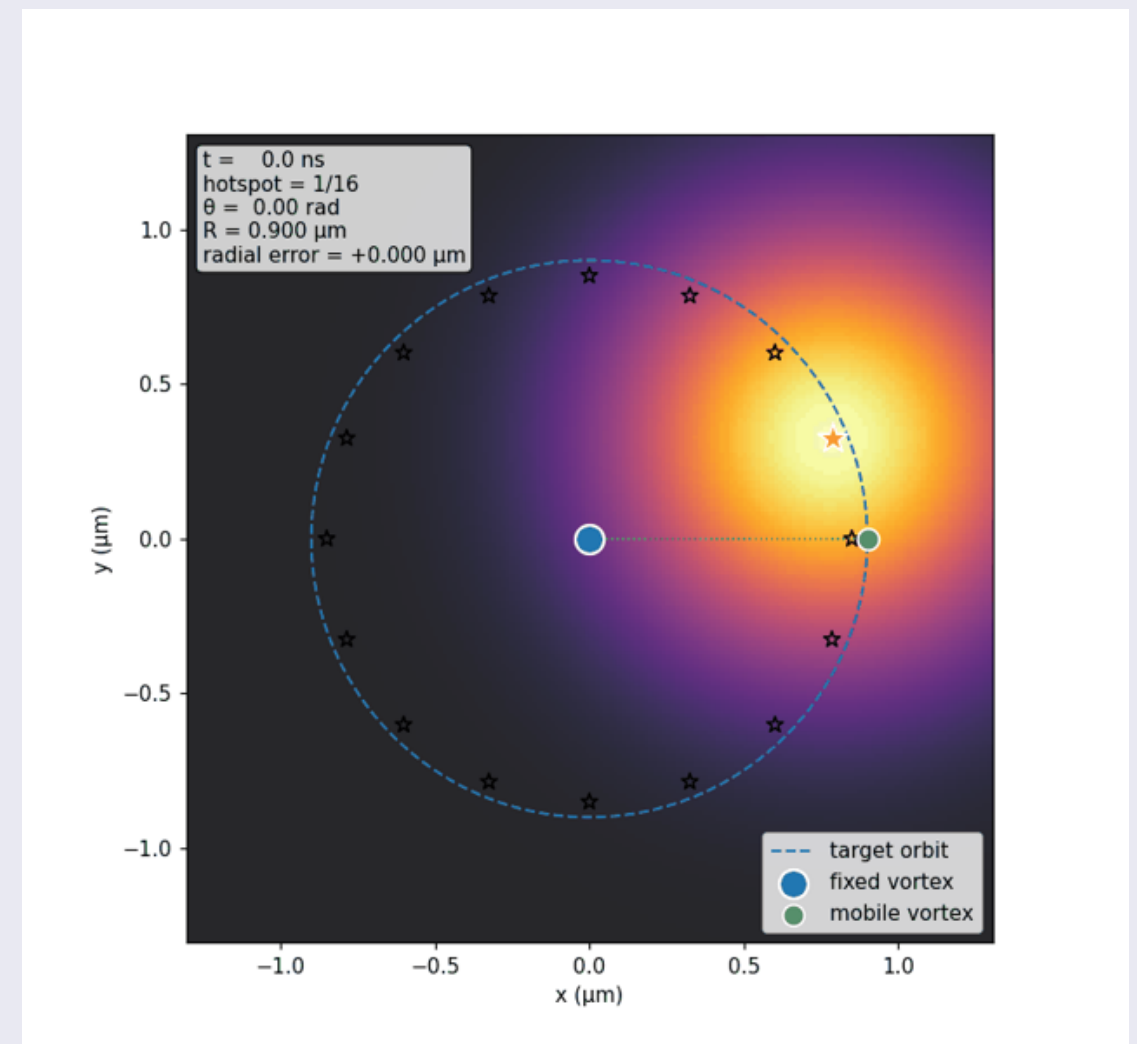
- Heating strength ΔT
- Distance between electrodes
- Switching time



A stronger temperature gradient is required when the distance between hotspots increases.

Programmable Hotspot Array

Multiple hotspots are arranged along a circular path and activated sequentially.



Using:

$$R_{orbit} = 0.90 \mu\text{m}$$

$$N = 16$$

the vortex follows a programmable circular trajectory.

Design Parameters

Response time

$$\tau = \frac{\eta \sigma_T^2}{\alpha_T \Delta T} \approx 7 \text{ ns}$$

A stable transfer requires:

$$t_{on} \gg \tau$$

and we choose:

$$t_{on} = 55 \text{ ns}$$

Hotspot spacing

The thermal force is maximized around:

$$d \sim \sigma_T$$

For circular transport:

$$R_{orbit} = 0.90 \mu\text{m}, \quad N = 16$$

giving:

$$d_{hot} = 0.345 \mu\text{m}$$

Conclusion and Future Work

Conclusion

- Developed a two-dimensional overdamped vortex dynamics model.
- Demonstrated:
 - vortex capture
 - vortex transfer
 - programmable circular motion
- Electrothermal manipulation provides a possible route toward programmable vortex control.

Future Work

- Solve realistic heat diffusion equation.
- Include stronger pinning effects.
- Compare with experimental parameters.

References

- J. Y. Ge et al., "Nanoscale assembly of superconducting vortices with scanning tunnelling microscope tip," Nature Communications 7, 13880 (2016).
- J. Bardeen and M. J. Stephen, "Theory of the motion of vortices in superconductors," Physical Review 140, A1197 (1965).
- S. Eley et al., "Glassy Dynamics in a heavy ion irradiated NbSe₂ crystal," Scientific Reports 8, 13162 (2018).
- I. Veshchunov et al., "Optical manipulation of single flux quanta," Nature Communications 7, 12801 (2016).